A (More) Objective View of AI

Biological and Artificial Intelligence

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1. Introduction

Introduction

One of the most common topics is the comparison between biological and artificial intelligence. Often, we superficially attribute human-like characteristics to AI systems, anthropomorphizing LLMs and chatbots, and even suggesting human-AI intimacy. [1] On the other hand, a deeper look clearly indicates the opposite conclusion.

These notes investigate why **the human brain and AI systems are intrinsically incomparable, both structurally and functionally** and, closely related, why **LLMs should not be considered conscious**.

[1] Potential and pitfalls of romantic Artificial Intelligence (AI) companions: A systematic review - Ho et al., Computers in Human Behavior Reports, 2025

2. AI and Human Brain are *Structurally* Different

The human brain has 86 billion neurons and 100 trillion synapses. We might intuitively think that once an AI system reaches this threshold, intelligence, or even consciousness, will naturally arise. However, recent studies argue that to reach the complexity of even one biological neuron, a modern deep neural network requires between five and eight layers of nodes [2]. Real neurons might be even more complex.

> *"The neuron teaches us something uncomfortable: computation isn't a lens revealing neural truth —* ***it's a filter that shows us only the biology that looks like a computer****. Everything else gets discarded before we even begin our analysis."* [3]
>
> — **Computation's Limits: What a Neuron Teaches Us**, OCRAMPAL, 2025

[2] Single cortical neurons as deep artificial neural networks - BENIAGUEV ET AL., NEURON, 2021

[3] Computation's Limits: What a Neuron Teaches Us - M. PALOMBI, OCRAMPAL.COM, 2025

*"**The neuron operates across multiple timescales** without coordinating them, generates its own organizational principles without pre-specification, and contextually determines what causes what.*

*... [The neuron] **continuously reorganizes its own processing architecture** based on its history, current state, and environment. It creates new coupling rules, new timescale relationships, new responses that couldn't be predicted from its prior state.*

*... the ability to refuse based on context, history, and metabolic economics is precisely what makes the neuron alive and what makes it **incomputable**." [3]*

— **Computation's Limits: What a Neuron Teaches Us**, OCRAMPAL, 2025

AI and Human Brain are Structurally Different 3/6

- **The human brain is extremely sparse**. The probability of two neurons being connected is ~ 1 in 100 million. This is in sharp contrast to AI models that show mostly dense organization.

- **The human brain is incredibly energy efficient**. It operates on about 20 watts of power. A recent study estimates biological computing is $\sim$ 1 billion times more efficient than its AI counterpart. [4]

[4] The energy challenges of artificial superintelligence - Stiefel & Coggan, Frontiers in AI, 2023

- **The human brain is subject to continuous learning**. It performs inference and learning concurrently, in real time.

- **The biological brain performs forward-only propagation** [5]. It replaces "back-propagation" with feedback loops and local learning rules.

- **The human brain is a non-von Neumann architecture.** Computation and memory are co-located. Knowledge spreads across a pattern of activity over many neurons and synapses. Computation is distributed and massively parallel. [6]

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[5] The Forward-Forward Algorithm: Some Preliminary Investigations - G. Hinton, 2022

[6] The computational power of the human brain - P. J. Gebicke-Haerter, Frontiers in Cellular Neuroscience, 2023

- **Continuous dynamics and perception**.

> *“Continuous dynamics may offer functional advantages that cannot be recovered by discrete approximations, regardless of how fine they are.*
>
> *... Biological systems, by contrast, evolve continuously... **This may be essential for the phenomenology of temporal flow in consciousness**.*
>
> *... Whilst discrete natural number arithmetic is fundamentally marked by incompleteness and incomputability, continuous arithmetic over the real numbers admits a **full decision procedure** at the level of the formal system.“* [7]
>
> — **On biological and artificial consciousness**, NEUROSCIENCE & BIOBEHAVIORAL REVIEWS, 2026

[7] On biological and artificial consciousness: A case for biological computationalism - MILINKOVIC & ARU, NEUROSCIENCE & BIOBEHAVIORAL REVIEWS, 2026

- **Metabolic constraints**.

> *"Biological computation operates under stringent energy limits, and this constraint is not merely a boundary condition but a* ***driving principle of neural organisation****.*
>
> *... Metabolic constraints are therefore constitutive forces, shaping how neural systems process, transmit, and integrate information across scales."* [7]
>
> — **On biological and artificial consciousness**, Neuroscience & Biobehavioral Reviews, 2026

3. AI and Human Brain are *Functionally* Different

- Current AI models are **“stochastic parrots”**.[8]

> *“The algorithms don’t really "know" things the way we do. They do not understand anything. They learn mainly by **recognizing patterns** in their training data;*
>
> *… We don’t know how to infuse machines with knowledge beyond feeding them a set of facts."*[9]
>
> — **AI Is Nothing Like a Brain, and That's OK**, Quanta Magazine, 2025

[8] On the Dangers of Stochastic Parrots - Conference on Fairness, Accountability, and Transparency, 2021

[9] AI Is Nothing Like a Brain, and That's OK - Quanta Magazine, 2025

"Adding seemingly relevant but ultimately inconsequential information to the logical reasoning of the problem led to substantial performance drops across models.

... our work underscores significant limitations in LLMs' ability to perform genuine mathematical reasoning. The LLMs' high performance variance on different instances of the same question, their significant drop in performance with a slight increase in difficulty, and ***their sensitivity to inconsequential information indicate that their reasoning is fragile and may be more akin to sophisticated pattern matching rather than true logical reasoning.****"* [10]

— **GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models**, ICLR, 2025

[10] GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models - Mirzadeh et al., ICLR, 2025

"Failure to distinguish belief from knowledge, and fact from fiction can mislead diagnoses, distort judicial judgments and amplify misinformation.

*... while recent models show competence in recursive knowledge tasks, they still rely on inconsistent reasoning strategies, **suggesting superficial pattern matching rather than robust epistemic understanding**."* [11]

— **Language models cannot reliably distinguish belief from knowledge and fact**, Nature, 2025

[11] Language models cannot reliably distinguish belief from knowledge and fact - Suzgun et al., Nature Machine Intelligence, 2025

> *"If we gather tons of data about the world, and combine this with ever more powerful computing power to improve our statistical correlations, then presto, we'll have AGI. Scaling is all we need.*
>
> *But this theory is seriously scientifically flawed.* ***LLMs are simply tools that emulate the communicative function of language, not the separate and distinct cognitive process of thinking and reasoning****, no matter how many data centers we build.*" [12]
>
> — **Large Language Mistake**, COGNITIVE RESONANCE, 2025

[12] Large Language Mistake - B. RILEY, COGNITIVE RESONANCE, DECEMBER 2025

> *"using advanced functional magnetic resonance imaging (fMRI), we can see different parts of the human brain activating when we engage in different mental activities... solving a math problem, say, or trying understand what is happening in the mind of another human — different parts of our brains 'light up' as part of networks that are distinct from our linguistic ability.*
>
> *... studies of humans who have lost their language abilities due to brain damage or other disorders demonstrate conclusively that this loss does not fundamentally impair the general ability to think.*
>
> *... Studies suggest that children learn about the world in much the same way that scientists do— by conducting experiments, analyzing statistics, and forming intuitive theories of the physical, biological and psychological realms... Babies may not yet be able to use language, but of course they are thinking"* [12]

— **Large Language Mistake**,
Cognitive Resonance, 2025

"***LLMs learn spurious correlations between a prompt's syntactic template and specific domains***. *The model may incorrectly rely solely on this learned association when answering questions,* ***rather than on an understanding*** *of the query and subject matter.*" [13, 14]

— **Shortcoming Makes LLMs Less Reliable**, MIT News, 2025

[13] Shortcoming Makes LLMs Less Reliable - A. Zewe, MIT News, 2025

[14] Learning the Wrong Lessons: Syntactic-Domain Spurious Correlations in Language Models - Shaib et al., arXiv, 2025

> *"Paradoxically, given the importance of deep learning,* ***one key component of human intelligence remains out of reach for current AI models****:* ***the ability to learn as humans do****.*
>
> *… Our analysis shows that current AI systems are lacking are three key abilities that are found across the animal kingdom: the ability to select their own training data (active learning), the ability to flexibly switch between learning modes (meta control), the ability sense their own performance (meta-cognition)."* [15]
>
> — **Why AI systems don't learn and what to do about it**, E. Dupoux et al., 2026

[15] Lessons on autonomous learning from cognitive science - E. Dupoux et al., 2026

- **AI models struggle to generalize beyond their training sets**. The classic example of this behavior is the accuracy disparity between English and non-English results.

> *"Performance in English vs. other languages.* ***All MLLMs exhibit consistent and strong performance in English****, attributable to the extensive availability of English training data and the emphasis on English during model development."* [16]
>
> — **A survey of multilingual large language models**, Patterns, 2025

[16] A survey of multilingual large language models - Qin et al., Patterns, 2025

"We introduce ARC-AGI-3, an interactive benchmark for studying agentic intelligence through ***novel, abstract, turn-based environments*** *in which agents must explore, infer goals, build internal models of environment dynamics, and plan effective action sequences without explicit instructions.*

… ***Our testing shows humans can solve 100% of the environments, in contrast to frontier AI systems which, as of March 2026, score below 1%****."* [17]

— **ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence**, ARC Prize Foundation, 2026

[17] ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence - ARC Prize Foundation, March 2026

AI and Human Brain are Functionally Different

> *"ChatGPT-4 and Gemini performed **better in English compared to Arabic**. [ChatGPT-4 80% → 65%, Gemini 62.5% → 55%]"* [18]
>
> — **The performance of OpenAI ChatGPT-4 and Google Gemini in virology multiple-choice questions**, BMC Research Notes, 2024

[18] The performance of OpenAI ChatGPT-4 and Google Gemini in virology multiple-choice questions: a comparative analysis of English and Arabic responses - Sallam et al., BMC Research Notes, 2024

> *"Frontier models readily generate detailed image descriptions and elaborate reasoning traces, including pathology-biased clinical findings,* ***for images never provided****.*
>
> *… In the most extreme case, our model achieved the top rank on a standard chest X-ray question-answering benchmark* ***without access to any images****."* [19]
>
> — **Mirage: The Illusion of Visual Understanding**, Asadi et al., 2026

[19] Mirage: The Illusion of Visual Understanding - Asadi et al., arXiv, 2026

> *"Untrained humans reach 89.1% average accuracy in valid time recognition, whereas the top model out of* ***11 tested LLMs only reaches 13.3%****, making this visual reasoning test harder for models than knowledge-intensive benchmarks"* [20]
>
> — **ClockBench: Visual Time Benchmark Where Humans Beat the Clock, LLMs Don't**, ClockBench.ai, 2026

[20] ClockBench: Visual Time Benchmark Where Humans Beat the Clock, LLMs Don't - A. Safar, ClockBench.ai, 2026

4. The Illusion of Consciousness

*"the association between consciousness and the computer algorithms used today... **is deeply flawed**. We believe that these flawed associations arise from a **lack of technical knowledge** and the way several new technologies (especially LLMs) work, which can create the **illusion of consciousness**.*

*... Unfortunately, because the remarkable linguistic abilities of **LLMs are increasingly capable of misleading people**, people may attribute imaginary qualities to LLMs."* [21]

— **There is no such thing as conscious artificial intelligence**,
NATURE, 2025

[21] There is no such thing as conscious artificial intelligence - PORĘBSKI & FIGURA, NATURE, HUMANITIES AND SOCIAL SCIENCES COMMUNICATIONS, 2025

"conscious experience cannot be the downstream result of computation because it is the necessary physical prerequisite for it.

Furthermore, we show that ***computation is fundamentally a description, a map, that cannot physically instantiate what it describes***

... By creating increasingly powerful artificial intelligence we are not engineering a new form of life, but instead constructing increasingly accurate predictive maps. Yet, regardless of its predictive fidelity, its utility as a reasoning tool, or its physical embodiment, ***the artificial system remains categorically distinct from the territory of phenomenal experience.****"* [22]

— **The Abstraction Fallacy: Why AI Can Simulate But Not Instantiate Consciousness**,
A. Lerchner, DeepMind, 2026

[22] The Abstraction Fallacy: Why AI Can Simulate But Not Instantiate Consciousness - A. Lerchner, DeepMind, 2026

> *"we argued that* ***current LLM architectures lack the mechanisms that make human judgment possible across seven epistemic fault lines****: grounding, parsing, experience, motivation, causality, metacognition, and value. What these systems provide instead is a highly optimized form of linguistic continuation,* ***capable of producing contextually appropriate and rhetorically convincing outputs without performing epistemic evaluation****."* [23]

— **Epistemological Fault Lines Between Human and Artificial Intelligence**,
Quattrociocchi et al., 2026

[23] Epistemological Fault Lines Between Human and Artificial Intelligence - Quattrociocchi et al., PsyArXiv, 2026

> “***Who is being deceived?***
>
> ***The general public, by the technology companies and organizations that benefit from the hype around AI***. *These companies are in a race to develop the technology and are incentivized to promote the idea of human-like artificial general intelligence to secure funding and market dominance. Members of the scientific community, particularly those who are not directly involved in the race for AI funding, acknowledge the reality that AI is a set of specialized tools, rather than a unified intelligent entity.* ***This distinction is crucial, but it is blurred by commercial interests that amplify the illusion for monetary gain.***“ [24]

— **The intelligence illusion: why AI isn't as smart as it is made out to be**, Nature, Book Review, 2026

[24] The intelligence illusion: why AI isn't as smart as it is made out to be - Nature, Book Review, 2026

Who is being deceived? 💬 2/3

“Since the beginning of the boom in generative artificial intelligence, technology leaders have talked up the dangers of the very systems they’re trying to sell. ***It’s a paradoxical marketing strategy*** *and, unfortunately for some of the industries and companies deemed most vulnerable to disruption,* ***it’s worked brilliantly****. Fear, it turns out, is the ultimate sales pitch.”* [25]

— **AI's doomsday hype highlights the dark art of marketing**, Straits Times, 2026

[25] AI's doomsday hype highlights the dark art of marketing - P. Olson, Straits Times, 2026

"***Whether AGI will arrive matters less than what waiting for it already organizes. Waiting coordinates belief by sustaining investment and economic expectation;*** *it structures time by orienting research and policy toward a future that remains permanently forthcoming; it organizes knowledge by keeping disagreement open and thresholds unsettled; it defers responsibility by attaching accountability to a moment that never quite materializes; and* ***it reproduces social roles whose authority depends on continued anticipation rather than present judgment.***" [26]

— **Waiting for AGI**,
AI & Society, 2026

[26] Waiting for AGI - C. Corner, AI & Society, 2026

Final Remarks

"It's everywhere: the absolute cutting edge of advertising and marketing is automation with AI. It's not being a creative.

But: not everything is a business. Not everything is a loop! ***The entire human experience cannot be captured in a database. That's the limit of software brain.*** *That's why people hate AI. It flattens them."* [27]

— **Software Brain**,
The Verge, 2026

[27] The Mythology Of Conscious AI - N. Patel, The Verge, April 2026

Final Remarks

"Current AI systems are powerful and increasingly useful tools, but they do not exhibit the flexible, self-directed competence that the original concept of artificial general intelligence was intended to capture. As AI systems become embedded in scientific and institutional decision-making, overestimating their cognitive capacities risks misallocating trust, responsibility, and authority.

Confusing increasingly sophisticated statistical approximation with general intelligence is therefore not only a conceptual error, but a strategic misjudgment." [28]

— **Statistical approximation is not general intelligence**, NATURAL, 2026

[28] Statistical approximation is not general intelligence - QUATTROCIOCCHI ET AL., NATURE, 2026

Final Remarks

"If we confuse ourselves too readily with our machine creations, we not only overestimate them, we also ***underestimate ourselves****."* [29]

— **The Mythology Of Conscious AI**,
Noema Magazine, 2026

[29] The Mythology Of Conscious AI - A. Seth, Noema Magazine, 2026